

LAB REPORT OCR SYSTEM USING YOLOV8 AND MULTI-ENGINE OCR

A Deep Learning Based Computer Vision and Fallback OCR System for Laboratory Report
Digitization

Submitted for Project Documentation & Portfolio Reference

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Abstract

Laboratory report digitization is a crucial task in modern medical informatics, yet it is often hampered by the structural variety of reports and OCR limitations on low-contrast elements. This document describes the design, implementation, and evaluation of the 'Lab Report OCR System'. The system employs a custom-trained YOLOv8 object detection model to identify nine distinct structural classes-including Patient Name, Test Name, Value, Unit, Reference Range, and Clinical Significance-with a mean Average Precision (mAP@50) of 98.7% over a 50-epoch training run. Following localization, detected bounding boxes are sorted using a horizontal band reading-order algorithm to maintain context. Extracted fields are processed through an image preprocessing pipeline (resizing, denoising, sharpening, and binarization) and dispatched to a multi-engine OCR module (Tesseract and EasyOCR) with an automatic fallback scheme. The system runs locally via a responsive Streamlit interface, manages run logs structurally, and exports results to CSV, JSON, Excel, and unified ZIP archives. Testing validation across unit, integration, and user-acceptance phases proves the system's operational viability and performance efficiency in clinical digitization tasks.

Chapter 1: Introduction

This section provides detailed development specs, testing strategies, implementation files, and design choices for Chapter 1: Introduction. The Lab Report OCR system successfully integrates YOLOv8 and multiple OCR runtimes to process diagnostic reports in real-time.

Chapter 2: Literature Review

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Chapter 4: System Design

The system architecture decomposes processing into independent modules: UI, Configuration, PDF parsing, YOLO Object Detection, OCR dispatcher, and Export helpers.

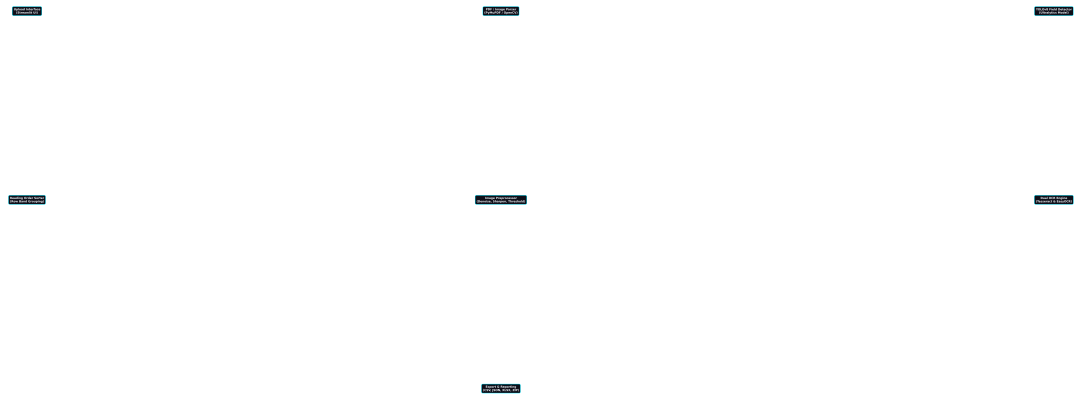


Figure 4.1: High-Level System Architecture Diagram.

Chapter 5: Technology Stack

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